R1.3 Energy from fuels

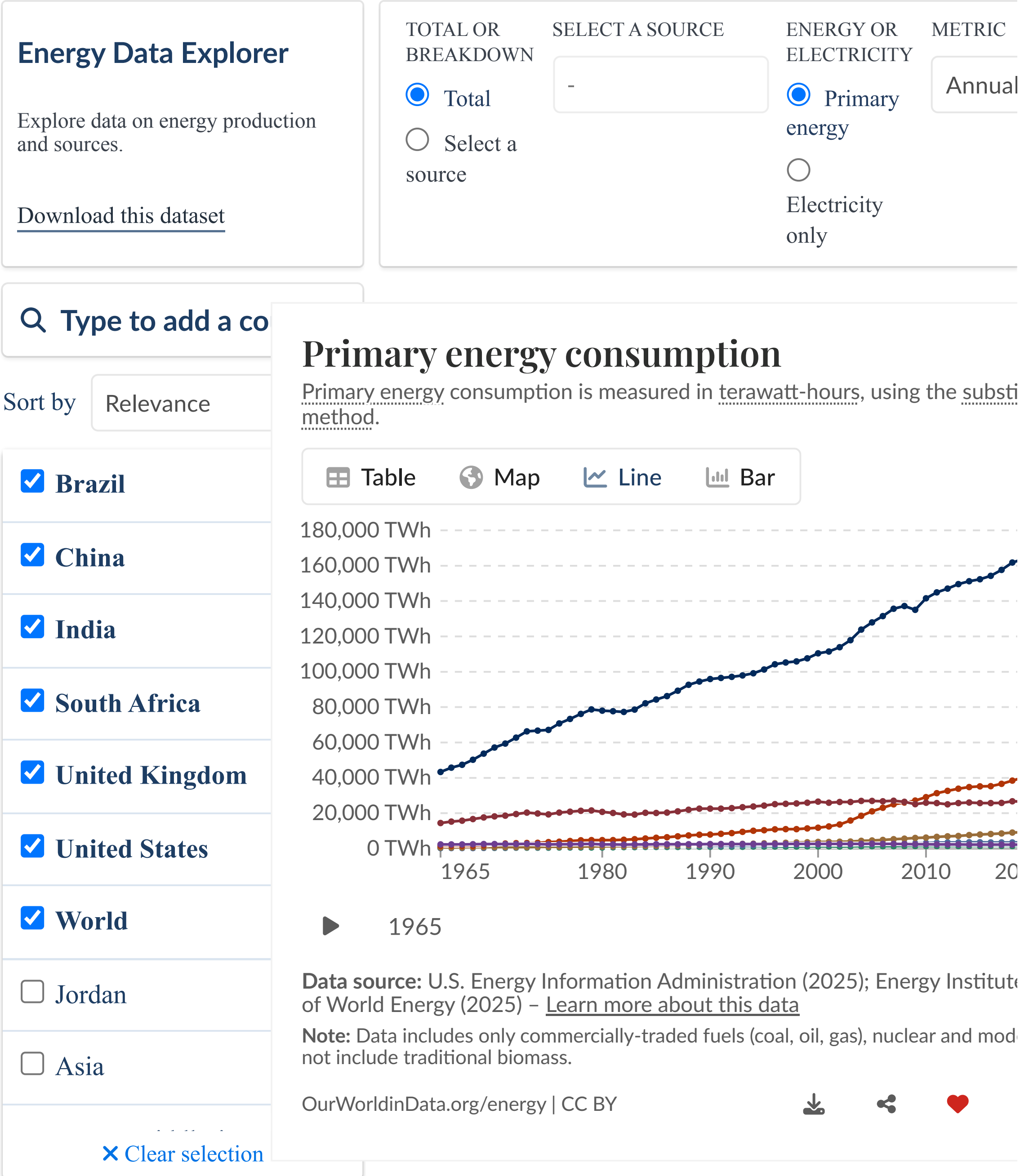
The big picture

Guiding question(s)

What are the challenges of using chemical energy to address our energy needs?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Over the past two hundred years humanity's needs for energy have increased at an unbelievable rate. The amount of energy needed today for our society to function is staggering in comparison to the energy needed in 1800. Some of this energy, like hydroelectricity and wind power, involves harnessing natural phenomena. Other types have a closer connection with chemistry, where we derive energy from chemical and nuclear reactions. In many situations chemical energy is still harnessed through the combustion of fossil fuels. Use **Interactive 1** to investigate the global demand for different fuels.



Interactive 1. Global demand for energy.

More information for interactive 1

The organic compounds in fossil fuels readily undergo chemical reactions to produce vast quantities of energy. Unfortunately, these reactions also produce harmful products that most scientists believe are having a negative impact on our global climate.

In the next section, you will learn about the energy and products involved in the reactions of fossil fuels as well as other types of reactants. You will also explore biofuels and fuel cells, which provide alternative methods of harnessing energy from chemical reactions.

Nature of Science

Aspect:

- Falsification
- Global impact of science

In January 2023, an article published in Science magazine raised awareness that major oil corporations had been suppressing data about the effects of fossil fuel combustion on the global climate since as early as 1977. The comprehensive article review states that 'For decades, some members of the fossil fuel industry tried to convince the public that a causative link between fossil fuel use and climate warming could not be made because the models used to project warming were too uncertain.'

https://www.science.org/doi/10.1126/science.abk0063 (https://www.science.org/doi/10.1126/science.abk0063)

This is even though the article also reports that ExxonMobil's average projected warming was 0.20 ± 0.04 °C per decade. This was found to agree with the projections of other independent academic and governmental agencies who published their findings between 1970 and 2007.

It is important to recognise that stakeholders can manipulate the data that is presented to the public. This misinformation, or distortion of data, can have far-reaching global impacts that we all need to be conscious of.

Prior learning

Before you study this subtopic make sure that you understand the following:

- Reactions are described as endothermic or exothermic, depending on the direction of energy transfer between the system and the surroundings (see section R1.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/endothermic-and-exothermic-reactions-id-45282/)).
- The relative stability of reactants and products determines whether reactions are endothermic or exothermic (see section R1.1.3 (https://app.kognity.com/:sectionlink:138726)).
- An oxidation state is a number assigned to an atom to show the number of electrons transferred in forming a bond. It is the charge that an atom would have if the compound were composed of ions (see section S3.1.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/)).